LEAST COST PATH USING DIGITAL ELEVATION MODEL (DEM): A CASE STUDY ON DARJEELING, WEST BENGAL

1. INTRODUCTION:

Geographic information systems (GIS) are frequently used by planners to facilitate land evaluation, which is a process of evaluating each piece of land in a given study area for a certain use and precedes a process of selecting one or more candidate sites for that use. If a raster-based GIS is used, land evaluation can be seen as ‘transformation of one or more sets of values associated with a grid of cells into another set of such values through a function reflecting one or more criteria’, which often takes numerical values within a set range. Planners may find a cost surface particularly useful if they seek a path and aim to minimize some kind of cost (or risk or any other undesirable consequence) associated with its location. This is because the search for a least-cost path on a cost surface can be easily cast as the problem of finding the shortest path in a graph by equating cells and pairs of adjacent cells with nodes and arcs, respectively, and there are efficient algorithms for its solution. A variety of applications of raster least-cost path models are found, some concerning right-of-ways for large-scale infrastructures such as highways, power lines and pipelines and others concerning conservation corridors for wildlife.

2. STUDY AREA:

The Darjeeling hill area is formed of comparatively recent rock structure that has a direct bearing on landslides. Heavy monsoon precipitation contributes to the landslides. Soils of Darjeeling hill areas are extremely varied, depending on elevation, degree of slope, vegetative cover and geolithology.

The Himalayas serve as the source of natural resources for the population residing in the hills as well as in the plains. As human population expands in the hills, forests are being depleted for the extension of agricultural lands, introduction of new settlements, roadways, etc. The growing changes coming in the wake of urbanisation and industrialisation leave deep impressions on the hill ecosystem.

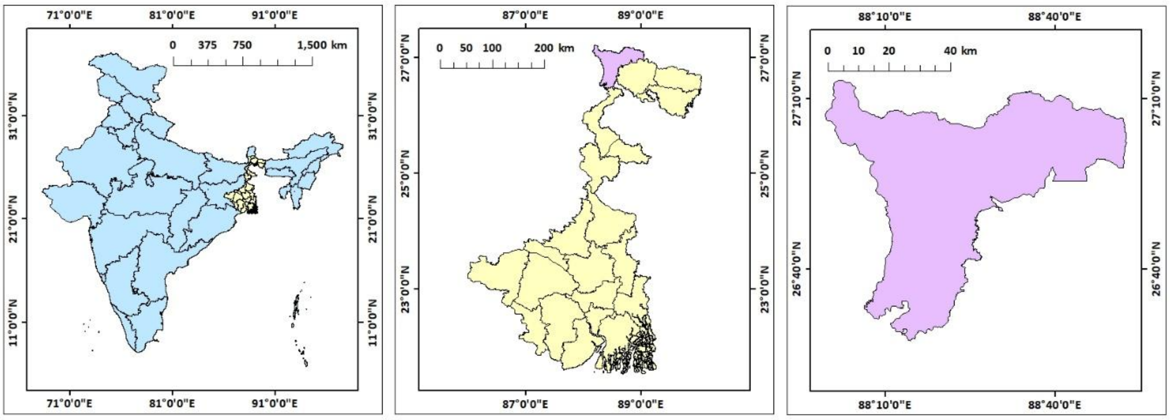

Figure 40: Location map of Darjeeling

The economy of Darjeeling hill area depends on tea production, horticulture, agriculture, forestry and tourism. The major portions of the forests are today found at elevations of 2,000 metres (6,600 ft) and above. The area in between 1,000 and 2,000 metres (3,300 and 6,600 ft) is cleared either for tea plantation or cultivation. About 30 percent of the forest covers found in the lower hills are deciduous. Evergreen forest constitutes only about 6 percent of the total forest coverage.

3. DATA SOURCES AND METHODOLOGY:

The following data is used for the present study –

DATA	SOURCE
Digital elevation model (DEM)	https://earthexplore r.usgs.gov

The digital elevation model (DEM) has been generated from the USGS website. Some important locations of Darjeeling have been saved as polygon shapefile and with the help of this DEM slope, cost distance, cost backlink and least cost path have been calculated between those locations using ArcGIS software.

4. RESULT AND DISCUSSION:

In the present study least cost paths have been generated from **Bagdogra international airport** to **Darjeeling mall**, **Kurseong**, **Mirik**, **Lamahatta** and **Tiger hill**.

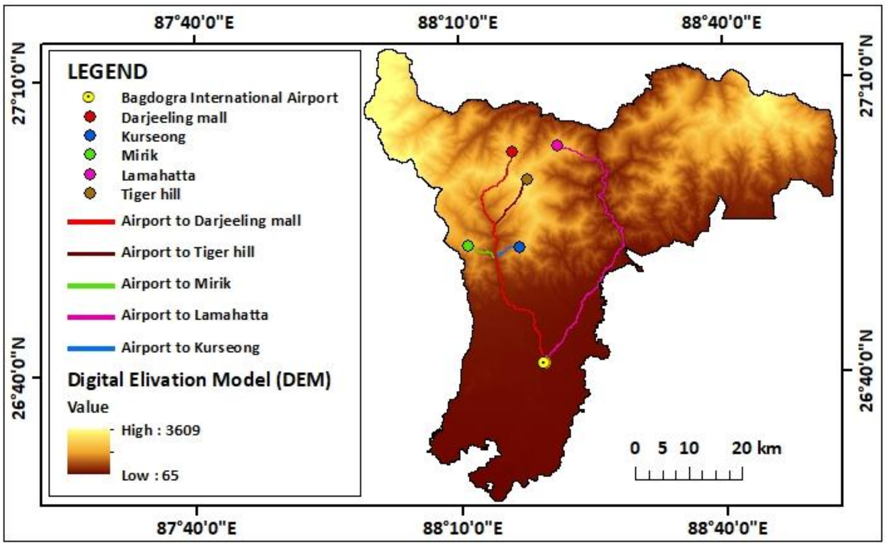

Figure 41: Least cost paths

5. CONCLUSION:

The present study proposes an effective method for calculating the least-cost path using digital elevation model (DEM), which addresses the challenges encountered in large-scale raster computation. Through a series of computational experiments, we demonstrate the effectiveness of this approach. By using several points and the Digital Elevation model (DEM) many analyses are done. Through which we can clearly understand the overall structure of Darjeeling. It can be also said that by observing these data several planning decisions could also be taken. Which could be also helpful for calculating 3D visualization of a terrain, watershed and viewshed analyses, and many other civil engineering, infrastructure, architectural and even defence and intelligence-related applications and uses.

CENTRALITY ANALYSIS: A CASE STUDY ON DUM DUM MUNICIPALITY, NORTH 24 PARGANA, WEST BENGAL

1. INTRODUCTION:

Centrality analysis is an essential concept in urban and regional planning, focusing on understanding spatial and social dynamics within a geographic area. It examines the distribution of key parameters such as population, infrastructure, and accessibility to identify areas of dominance or importance within a city or municipality. This approach is particularly significant in the context of Dum Dum Municipality, an urban area in West Bengal, India, characterized by its dense population, diverse economic activities, and well-connected road network.

The study of centrality in Dum Dum Municipality involves analysing various parameters, including the literate population, working population, total population, and road density. These indicators collectively provide insights into the functional importance of different locations within the municipality. Literacy rates are a measure of human capital, reflecting the socio-economic potential of the area, while the working population indicates economic engagement and the availability of labour. Total population density serves as a fundamental indicator of urban congestion and service demand. Additionally, road density is a critical determinant of connectivity, accessibility, and mobility, influencing the centrality of specific zones.

Centrality analysis has practical implications for urban management, such as optimizing public service delivery, identifying marginal and underserved areas, and planning for sustainable urban growth. By integrating spatial data and demographic parameters, this analysis can reveal the hierarchy of locations, from highly central zones with abundant resources and services to marginal areas requiring policy intervention. For instance, centrality indices can identify regions with high accessibility to schools, workplaces, and healthcare facilities, providing a framework for equitable urban planning.

In the case of Dum Dum Municipality, the visualization of centrality using tools like GIS enables planners to map and interpret spatial patterns effectively. Overlaying demographic data with infrastructure elements, such as roads, highlights the interplay between human activity and the urban fabric. This study aims to contribute to better governance and resource allocation by identifying core urban zones and their interactions with peripheral areas. Centrality analysis thus serves as a foundation for creating inclusive and balanced urban spaces that cater to the diverse needs of the population.

2. STUDY AREA:

Dum Dum Municipality, located in the North 24 Parganas district of West Bengal, India, is a densely populated and urbanized region with significant socio-economic and historical importance. The municipality lies between approximately 22°37'N to 22°39'N latitude and 88°23'E to 88°26'E longitude, covering an area of around 9.5 square kilometers. Dum Dum serves as a vital suburban hub within the Kolkata Metropolitan Area, with well-established residential, industrial, and commercial zones.

The region is characterized by its strategic location and connectivity, particularly due to the presence of Netaji Subhash Chandra Bose International Airport, one of the busiest airports in India. Additionally, Dum Dum is intersected by major roadways and railway lines, facilitating seamless transportation and economic activities. The municipality is divided into several wards, each exhibiting distinct socio-demographic and infrastructural features. The population density of Dum Dum is notably high, reflecting the region’s urban sprawl and increasing demand for housing, infrastructure, and public services.

Dum Dum is also known for its cultural diversity and historical significance, being one of the oldest settlements near Kolkata. The area hosts a mix of traditional neighborhoods and modern residential complexes, showcasing a blend of heritage and contemporary development. Key parameters such as literacy rate, working population, and road density reflect the urban vibrancy and functional dynamics of the region.

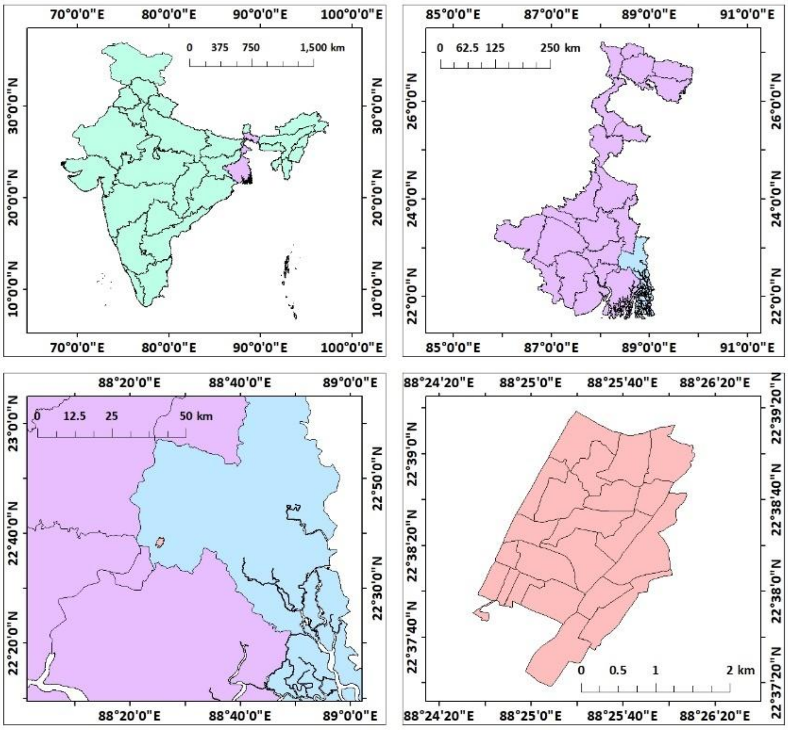

Figure 101: Location map of Dum Dum municipality

The municipality experiences a humid subtropical climate, with heavy rainfall during the monsoon season, influencing its urban planning and infrastructural resilience. The socio-economic fabric of Dum Dum, coupled with its infrastructural attributes, makes it an ideal study area for centrality analysis. Understanding the distribution of resources and accessibility within Dum Dum can provide critical insights for sustainable urban planning, equitable development, and improved quality of life for its residents.

3. DATA SOURCES AND METHODOLOGY:

DATA SOURCES:

The data utilized for this study was primarily sourced from the **Census of India 2011**, which provides comprehensive demographic, socio-economic, and infrastructural statistics at various geographic levels. The census data offers a rich dataset on parameters such as total population, literacy rate, and workforce participation, all of which were essential for analysing the centrality of Dum Dum Municipality.

METHODOLOGY:

The methodology for this study involved using ArcGIS software for geospatial analysis to evaluate the centrality indices of Dum Dum Municipality based on selected parameters. The entire process was designed to combine demographic data from the Census of India 2011 with spatial data to derive insights into the functional hierarchy and accessibility patterns of the area.

Initially, demographic and socio-economic data for Dum Dum Municipality, including total population, literacy rates, and the working population, were extracted from the Census 2011 dataset. This data was organized and digitized into attribute tables for integration into a Geographic Information System (GIS) environment. Road network data was acquired and mapped to calculate road density, a critical indicator of urban connectivity and accessibility.

TABLE FOR CALCULATION OF CI

OBJECTID	NEAR_FID	DISTANCE_WEI	DISTANCE_POP	DISTANCE_LIT	DISTANCE_WORK	DISTANCE_ROAD	AVERAGE	CI
1	20	1537.243156	1676.687056	1688.760702	1714.773462	1610.589006	1645.610677	4.588277
2	21	1080.799535	1185.5156	1194.54588	1213.294	1060.834403	1146.997883	3.830605
3	19	889.6097195	1031.078331	1043.451338	1070.427296	995.1814493	1005.949627	3.587354
4	17	1200.623386	1320.758711	1331.82296	1356.859791	1356.436343	1313.300238	4.098907
5	12	833.611869	948.0918545	958.0263365	978.8986508	834.9140978	910.7085617	3.413311
6	16	1194.890365	1303.375162	1313.603266	1337.061044	1357.669689	1301.319905	4.080169
7	18	973.6520843	1101.517919	1113.13931	1139.186564	1122.487624	1089.9967	3.73421
8	14	544.6685943	670.1755135	681.7398193	707.714307	699.2601137	660.7116696	2.907315
9	13	269.0149232	395.2509465	406.484778	430.5600802	317.9068976	363.8435251	2.157463
10	15	891.4107347	975.8769944	984.4426557	1004.690636	1058.84782	983.0537682	3.546294
11	11	873.9103047	856.6487837	855.4899452	851.8039508	721.1544622	831.8014893	3.26209
12	10	557.556775	520.8335471	518.5950221	512.9433374	394.6925623	500.9242488	2.531466
13	8	609.1051366	646.3030338	651.6028859	665.4102101	764.5738934	667.399032	2.921991
14	9	235.2965351	104.1158487	93.6352114	71.56939483	158.5169591	132.6267898	1.302572
15	6	567.7667409	469.7540243	462.8644119	450.1659799	612.0186196	512.5139553	2.560583
16	5	705.5414287	575.7799873	564.3803348	538.8912271	577.3408603	592.3867676	2.752889
17	4	1081.460199	977.4748446	968.3086953	947.3097228	922.0857681	979.327846	3.539567
18	7	824.4810986	743.8175046	738.2929069	728.4299791	889.6602518	784.9363482	3.168862
19	1	957.3029159	816.9620629	804.8309197	778.7220514	894.8290053	850.529391	3.298608
20	2	1230.959864	1093.454837	1081.306313	1054.426909	1115.861029	1115.20179	3.777138
21	3	1555.220837	1430.436643	1419.318344	1394.299966	1412.146227	1442.284403	4.295478
22	0	1315.293661	1180.306973	1168.830583	1144.504336	1276.601619	1217.107434	3.945941

Administrative boundary shapefiles, including ward and municipal boundaries, were imported into ArcGIS to provide spatial context. The road network was digitized, and its density was calculated by dividing the total length of roads by the area of each ward. Spatial joins and overlays were conducted to associate the demographic attributes with corresponding wards.

To compute centrality indices, buffer zones were created around central points, such as major roads and administrative centres, at different distances. This helped identify zones of varying centrality, from highly central to marginal locations. Weighted overlays were conducted, combining parameters like population density, literacy rate, and road density to calculate composite centrality scores. Each parameter was normalized to ensure comparability before assigning weights based on its significance in influencing centrality. The spatial outputs were represented as thematic maps using ArcGIS, with distinct colour schemes for different centrality levels. This visualization helped identify highly central areas, typically characterized by higher accessibility, road density, and socio-economic activity, and marginal areas requiring policy attention. The maps were validated by comparing them with existing municipal data and field observations.

4. RESULTS AND DISCUSSION:

The centrality analysis of Dum Dum Municipality, based on the selected parameters—literacy rate, working population, total population, and road density—has provided insightful results regarding the spatial distribution of socio-economic and infrastructural features across the region. Using ArcGIS software, these parameters were mapped and analyzed to calculate the centrality index for different areas within the municipality.

The centrality index was computed based on a weighted overlay of the four parameters, with a focus on how these factors influence the accessibility and economic activity in different parts of Dum Dum. The results, displayed in the form of thematic maps, show a clear spatial variation in the centrality index across the municipality, reflecting the influence of socio-economic activities and urban infrastructure on the overall accessibility of the area.

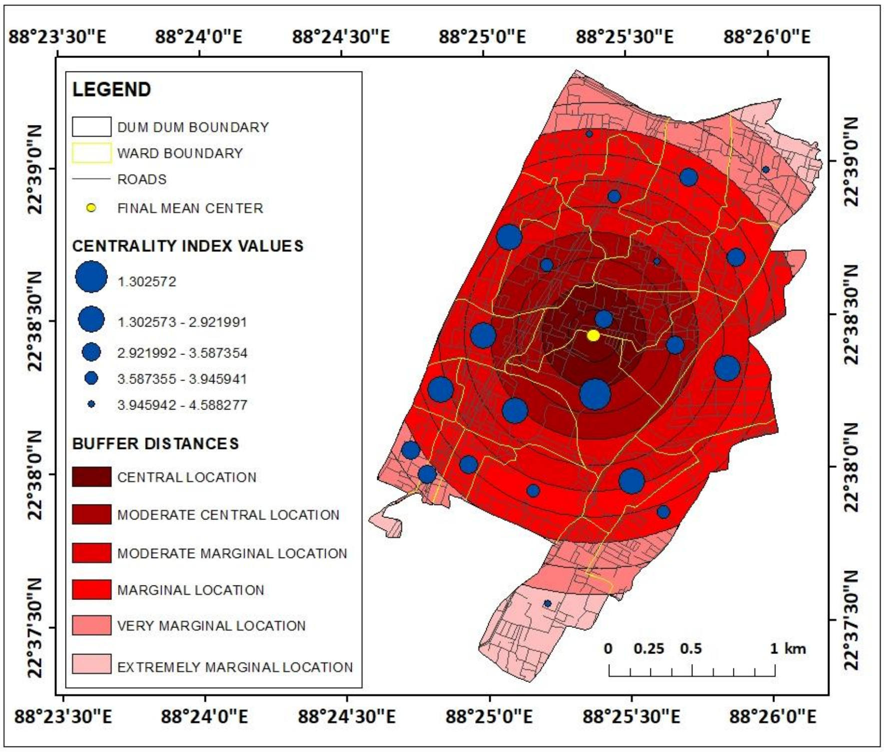

Figure 102: Centrality map on Dum Dum municipality

The central regions of Dum Dum, particularly around the core commercial and residential areas, exhibited the highest centrality index values. These areas are characterized by higher population densities, greater working populations, and increased literacy rates, which are typical of urban centers with better infrastructure. The high road density in these areas facilitates easy accessibility and movement, thereby reinforcing their centrality. These regions also align with the proximity to major transportation hubs such as the Dum Dum Junction and the Netaji Subhash Chandra Bose International Airport, contributing to their central position in terms of both connectivity and socio-economic activities.

In these highly central areas, the availability of services and facilities such as healthcare, education, and commerce are significantly better than in the outer wards. The integration of road infrastructure, coupled with the dense concentration of residents and workers, makes these regions economically vibrant and key drivers of urban growth.

Moving away from the core, areas exhibiting moderate centrality are often characterized by a balance between population density and road infrastructure. These regions may have adequate road connectivity but lack the same concentration of services and economic activities as the central areas. In these regions, the working population is still significant, and literacy rates are relatively high, but the overall urban experience is less dense compared to the core areas. The availability of public services and infrastructure like markets, schools, and healthcare is generally accessible, but not as concentrated as in highly central zones.

Moderately central locations are typically areas undergoing urban development, where there is room for growth in terms of infrastructure and services. These areas may benefit from future urban planning initiatives focused on enhancing connectivity and boosting socio-economic activities, improving their centrality in the long term.

The marginal locations, as identified in the results, are located in the periphery of the municipality. These areas are often sparsely populated and lack adequate road infrastructure, which limits their accessibility and socio-economic opportunities. In terms of population, literacy rates, and workforce participation, these regions tend to be less active, with fewer resources available. The low road density in these areas further exacerbates their marginality, making it harder for residents to access essential services and for businesses to thrive. These areas often face challenges such as inadequate public transportation and limited infrastructure, contributing to their lower centrality index.

The very marginal locations, primarily on the outskirts of Dum Dum, represent the areas with the lowest centrality index values. These regions are characterized by a very low concentration of population, limited road network, and low literacy rates. Often underdeveloped, these areas lack basic infrastructure and services, hindering economic activities and social mobility. As a result, they experience challenges in terms of accessibility to education, healthcare, and employment opportunities.

The spatial distribution of these marginal and very marginal locations highlights a critical need for targeted policy interventions. Urban planners and policymakers could focus on improving road connectivity, enhancing educational and employment opportunities, and investing in infrastructure to bring these areas closer to the centrality of the more developed regions. Strategies such as improving transportation networks, increasing the number of community services, and encouraging private sector investments in infrastructure could help elevate the centrality index of these areas over time.

5. CONSLUSION:

The centrality analysis of Dum Dum Municipality has provided valuable insights into the socio-economic and infrastructural characteristics that shape the accessibility and development of different areas within the region. By utilizing parameters such as population density, literacy rate, working population, and road density, the study has mapped out the spatial variations in centrality across the municipality. The results indicate that the central areas, characterized by high population density and strong infrastructure, exhibit the highest centrality index, making them the key hubs of economic and social activity. In contrast, peripheral areas with lower infrastructure and population density have a significantly lower centrality, reflecting challenges in accessibility and service provision.

These findings underline the importance of targeted urban planning and infrastructure development in marginalized areas. Addressing the gaps in road connectivity, enhancing services, and improving socio-economic conditions in the outer regions can help reduce spatial inequalities and foster more balanced urban growth. By prioritizing inclusive development strategies, Dum Dum Municipality can work towards creating a more equitable urban environment, where all areas benefit from improved accessibility and socio-economic opportunities. The use of ArcGIS software in this analysis has proven to be an effective tool in visualizing and understanding the spatial dynamics of urban centrality, providing a basis for future urban planning interventions.

PREPARATION OF WEB GIS ON SOUTH BENGAL TOURISM USING GOOGLE API

1. INTRODUCTION:

Web GIS (also known as Web-Based GIS), or Web Geographic Information Systems, are GIS that employ the World Wide Web to facilitate the storage, visualization, analysis, and distribution of spatial information over the Internet. The World Wide Web, or the Web, is an information system that uses the internet to host, share, and distribute documents, images, and other data. Web GIS involves using the World Wide Web to facilitate GIS tasks traditionally done on a desktop computer, as well as enabling the sharing of maps and spatial data. While Web GIS and Internet GIS are sometimes used interchangeably, they are different concepts. Web GIS is a subset of Internet GIS, which is itself a subset of distributed GIS, which itself is a subset of broader Geographic information system. The most common application of Web GIS is Web mapping, so much so that the two terms are often used interchangeably in much the same way as Digital mapping and GIS. However, Web GIS and web mapping are distinct concepts, with web mapping not necessarily requiring a Web GIS.

2. STUDY AREA:

The southern part of West Bengal, India, encompasses a wide area, including districts like South 24 Parganas, Purba Medinipur, Paschim Medinipur, Howrah, Hooghly. Nadia and Kolkata. Here are approximate coordinates for the central area of this region, which can serve as a reference point: Latitude: 22.50° N to 21.50° N, Longitude: 87.00° E to 88.50° E. These coordinates roughly cover the heart of the southern region of West Bengal, including major cities like Kolkata and significant natural

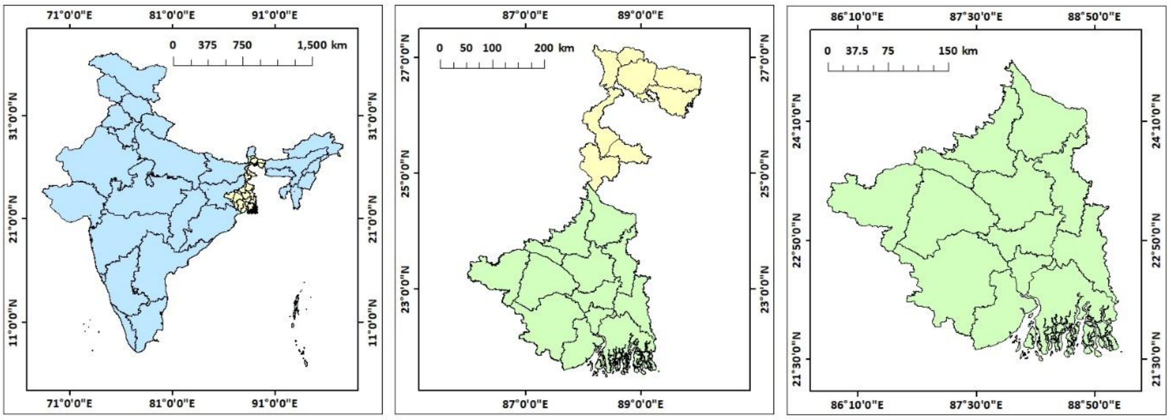

Figure 38: Location map of southern part of West Bengal

landmarks such as the Sundarbans mangrove forest. In the context of South Bengal as the study area, Web GIS for tourism map preparation emerges as a pivotal strategy to harness the region's rich cultural, historical, and natural assets. By employing Web GIS tools, we can craft dynamic maps that vividly depict the diverse array of attractions spanning from the verdant Sundarbans mangrove forests to the vibrant cultural heritage sites in cities like Kolkata. These maps serve as invaluable resources for tourists, offering detailed insights into accommodation options, transportation networks, and points of interest, thereby facilitating seamless exploration and enhancing overall visitor satisfaction.

3. DATA SOURCES AND METHODOLOGY:

The locations of the tourist spots of the south Bengal has been extracted with the help of Google Earth Pro and saved as Kml layer. Using those locations the following work has been done.

4. RESULT AND DISCUSSION:

Link for Web GIS on South Bengal tourism is given below –

https://www.google.com/maps/d/u/0/edit?mid=1NETXAVc4QYuns5kfWLsGDbBnMUEGXaE&ll=24.744192186842938%2C87.47832650000002&z=7

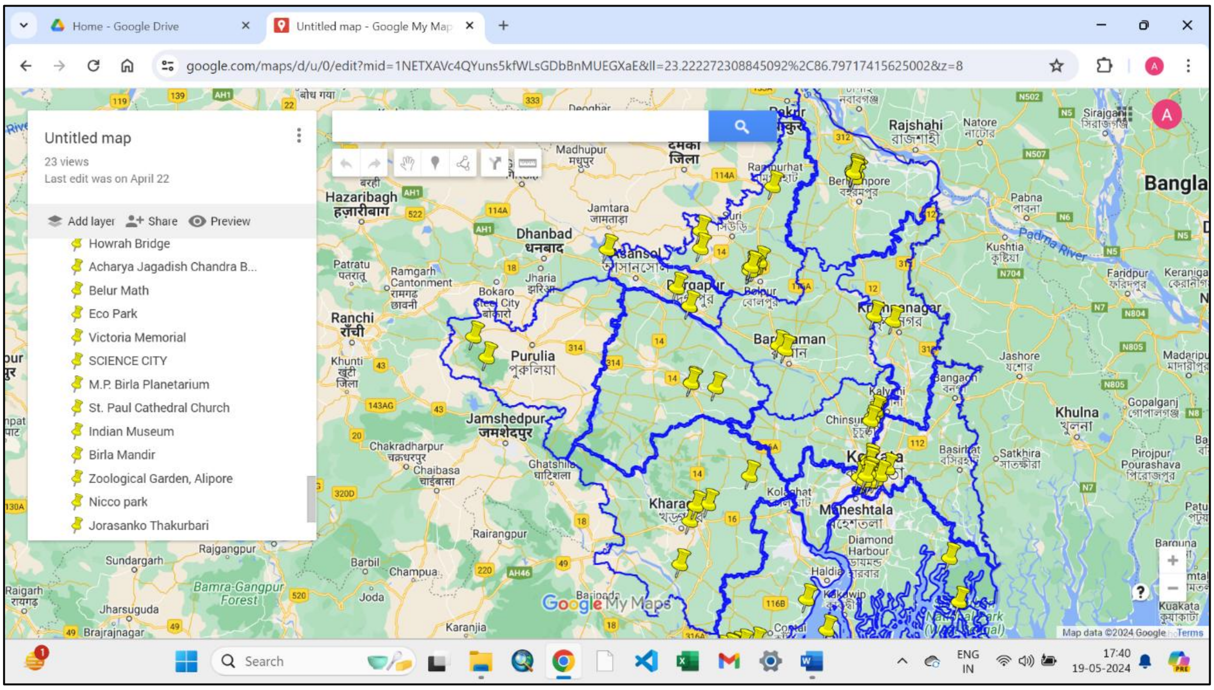

Figure 39: Screenshot of Web GIS

5. CONCLUSION:

Utilizing Web GIS for tourism map preparation offers a dynamic and accessible platform to showcase diverse attractions, facilitating enhanced visitor experiences and strategic planning. Through the integration of spatial data and interactive features, such maps empower tourists with comprehensive information, from points of interest to navigation tools, fostering informed decision-making and exploration. Furthermore, the collaborative nature of Web GIS enables continual updates and customization, ensuring the map remains current and tailored to evolving tourist needs, thereby promoting sustainable tourism development and fostering economic growth within the region.